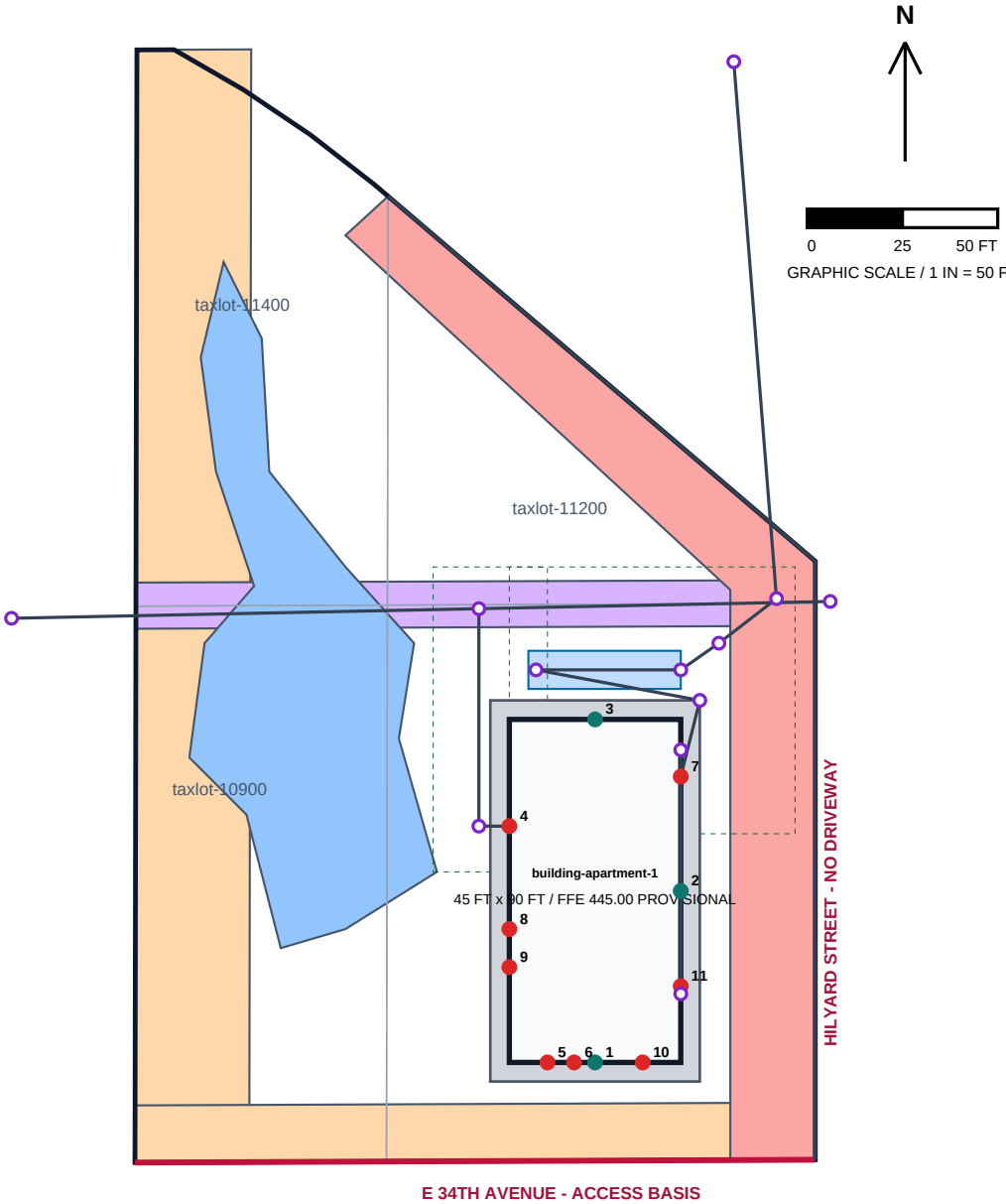


MODEL STUDIO - HILYARD SITE LAYOUT / BUILDING INTERFACES

Product-development test data only - no survey, engineering, capacity, approval, or code-compliance claim



BASIS / CONTROL

Horizontal: EPSG:6823, international feet
DRAWING GRID ORIGIN: E 186225.00 / N 98450.00
Vertical: NAVD88 feet; benchmark and elevations UNKNOWN
Taxlots: City GIS reference-derived; NOT A SURVEY
Replace taxlot geometry first when a boundary survey arrives
Constraints: dimensioned/digitized reference geometry
Output parity tolerance: 0.01 ft (not source accuracy)
Utility routing is on discipline sheets; capacity/approval unknown

BUILDING INTERFACES - PERMANENT TERMINAL IDS

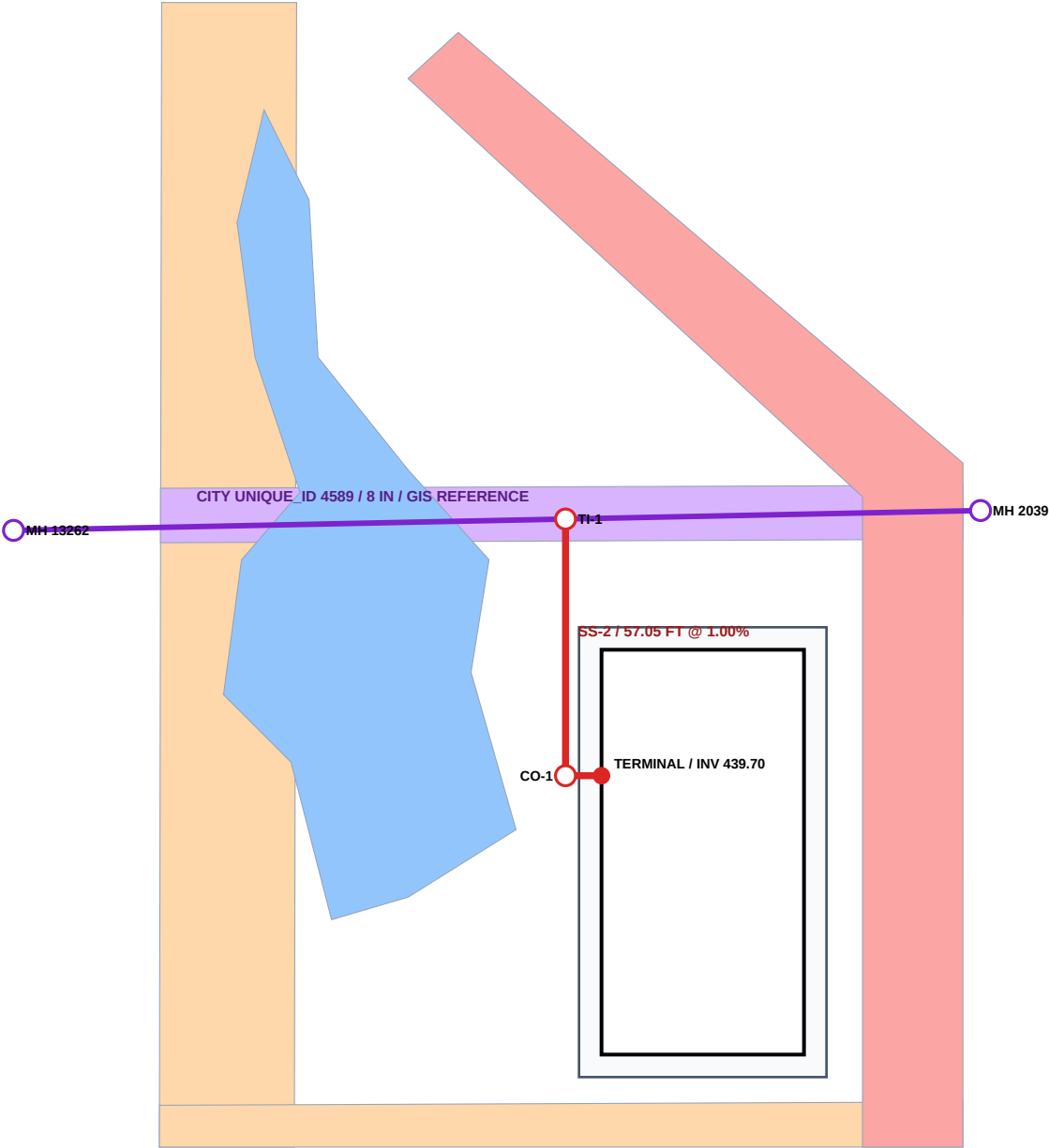
- 01 entry-south-primary [pedestrian]
- 02 entry-east-service [service]
- 03 entry-north-pedestrian [pedestrian]
- 04 penetration-sanitary [sanitary]
- 05 penetration-domestic-water [domestic_water]
- 06 penetration-fire-water [fire_water]
- 07 penetration-roof-drainage [roof_drainage]
- 08 penetration-electric [electric]
- 09 penetration-telecom-fiber [telecom_fiber]
- 10 penetration-gas [gas]
- 11 penetration-site-lighting [site_lighting]

CONSTRAINT LEGEND

- constraint-eweb-west
- constraint-eweb-south
- constraint-sanitary-easement
- constraint-row-dedication
- constraint-wetland

MODEL STUDIO - HILYARD SANITARY SEWER PLAN

Reference GIS context plus reviewed-assumption fictional service; no survey, capacity, tie-in approval, or final grading claim



SANITARY BASIS

City main: UNIQUE_ID 4589
8 IN / folded-form liner / reference GIS
GIS inverts: 438.87 -> 437.47 NAVD88
Proposed service: 6 IN PVC SDR35
SS-1: 8.00 FT @ 1.00%
SS-2: 57.05 FT @ 1.00%
TI-1 incoming INV 439.05 (assumed)
Public main INV at TI-1: 438.07 (interpolated)
Minimum cover checked: 4.00 FT
All proposed cover <5 FT: shallow review warning
Tie-in approval and capacity: UNKNOWN
Surface/FFE: PROVISIONAL; not survey control

FIELD WORKFLOW

- 1 Pothole/survey existing main
- 2 Obtain City tie and capacity acceptance
- 3 Verify FFE, terminal, and finished grade
- 4 Stake line/grade; inspect bedding
- 5 Test, inspect, backfill/compact
- 6 Capture as-built coordinates/inverts

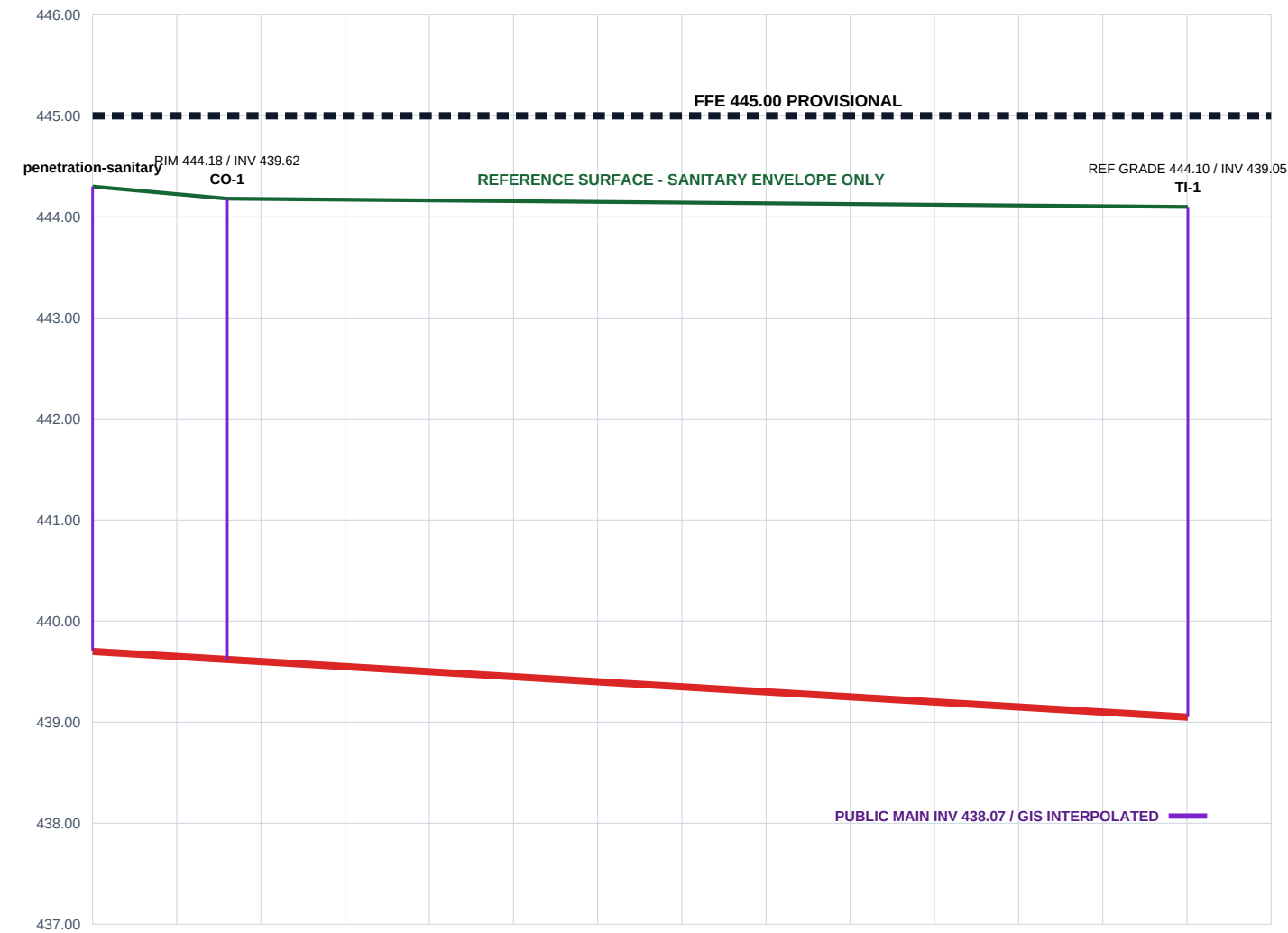
STABLE ASSET IDS

sanitary-service-seg-01
sanitary-service-seg-02
sanitary-cleanout-01
sanitary-tie-in-01
sanitary-public-main-4589-upstream
sanitary-public-main-4589-downstream

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD SANITARY SEWER PROFILE

Stations follow penetration-sanitary to TI-1; all proposed vertical values are provisional test assumptions

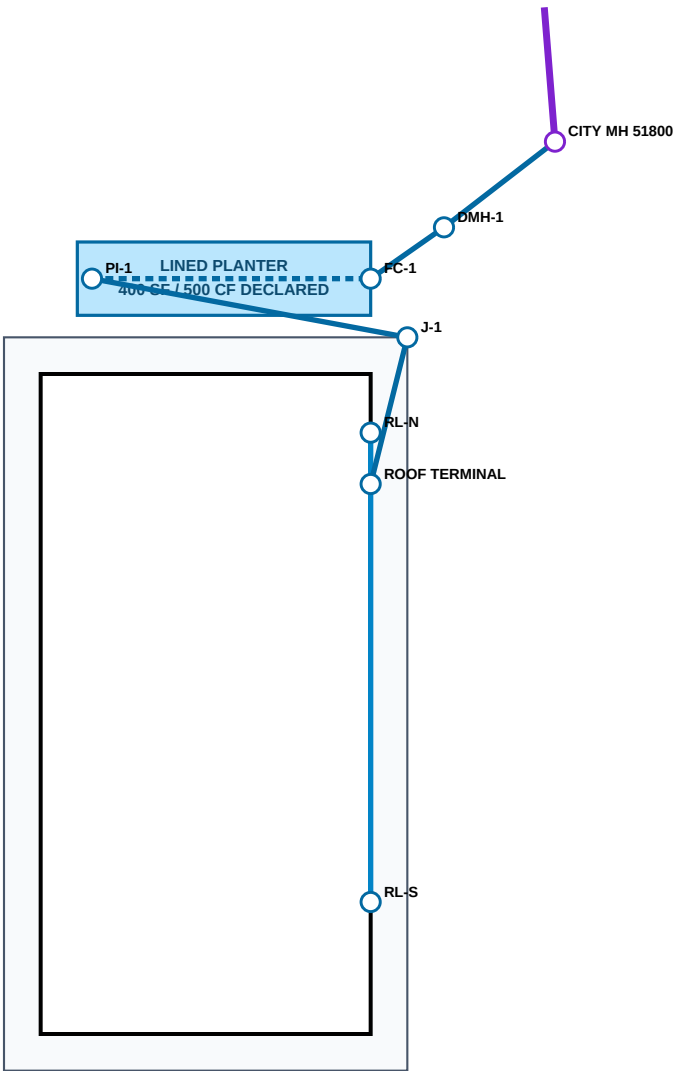


SEGMENT	STATIONS	LENGTH	SLOPE	INVERTS NAVD88	COVER	MATERIAL
san-edge-service-01	0+00.00 - 0+08.00	8.00 FT	1.00%	439.70 -> 439.62	4.06-4.10 FT	6 IN PVC SDR35
san-edge-service-02	0+08.00 - 0+65.05	57.05 FT	1.00%	439.62 -> 439.05	4.06-4.55 FT	6 IN PVC SDR35

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD STORM / ROOF DRAINAGE PLAN

Roof-only fictional treatment/conveyance system; final grading, capacity, HGL, infiltration, outfall approval, and survey are unresolved



HYDROLOGIC / VERTICAL BASIS

Roof area: 4,050 SF / C = 1.00 (assumed)
WQ storm: 1.40 IN / screen volume 472.50 CF
10-year storm: 4.46 IN / unrouted volume 1,505.25 CF
Planter declared surface storage: 500.00 CF
Method: rainfall-volume screening only; NOT ROUTED
Reference surface / FFE 445.00: PROVISIONAL
Public main: City GIS UNIQUE_ID 4183 / 21 IN
MH 51800 rim 444.87 / public out INV 434.33
Capacity, HGL, tie/outfall approval: UNKNOWN
Infiltration/geotechnical suitability: UNKNOWN

ALIGNMENT / CONFLICT

Private roof collector: 6 IN PVC
Facility underdrain: 6 IN perforated PVC
Outlet/public connection: 10 IN PVC SDR35
DMH-1 external drop: 3.95 FT (assumed)
Storm below sanitary crossing clearance: 1.253 FT
Reviewed minimum clearance: 1.00 FT
Pothole/survey both utilities before real design
No catch basin/site runoff: final grading excluded

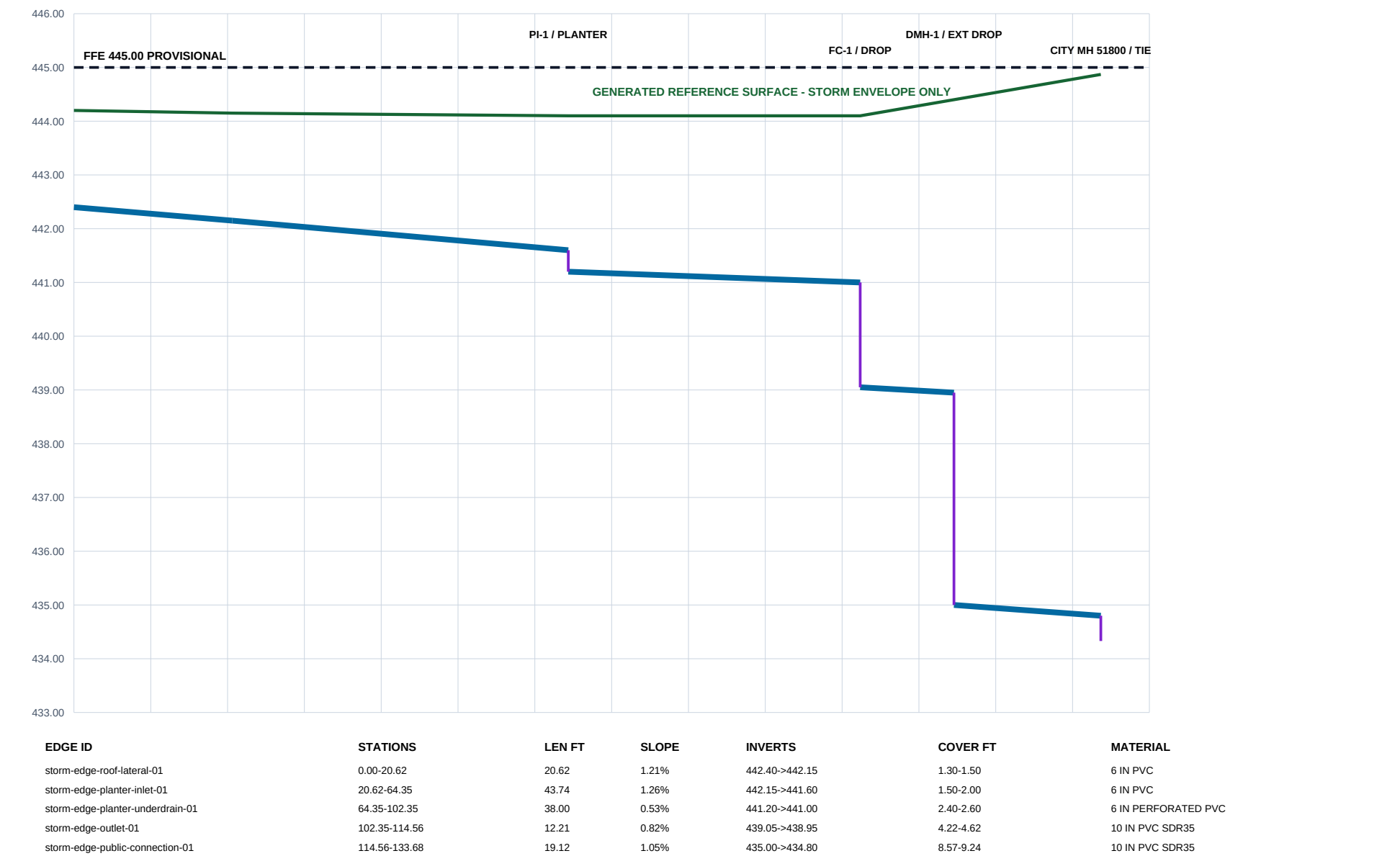
FIELD HOLD POINTS

- 1 Confirm survey/control and final grades
- 2 Verify roof drains/overflow with architect
- 3 Obtain geotechnical infiltration testing
- 4 Confirm public capacity/HGL and City tie approval
- 5 Pothole sanitary/storm crossing
- 6 Inspect liner, media, bedding, tests, as-builts

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD STORM / ROOF DRAINAGE PROFILE

Stations run from permanent roof terminal through planter/flow control/drop manhole to assumed City tie; elevations are test-basis values



FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

FICTIONAL — TEST DATA — NOT FOR CONSTRUCTION

MODEL STUDIO - HILYARD STORM ASSET SCHEDULE / TEST DETAILS

Stable IDs, numeric test basis, field hold points, and explicit unknowns from the canonical semantic model

CONVEYANCE ASSET SCHEDULE

ASSET ID	TYPE	DIA	LEN	SLOPE	INVERTS NAVD88	STATUS
storm-edge-roof-lateral-01	private_roof_lateral	6 IN	20.62	1.21%	442.40->442.15	ASSUMED / GENERATED
storm-edge-planter-inlet-01	private_facility_lateral	6 IN	43.74	1.26%	442.15->441.60	ASSUMED / GENERATED
storm-edge-planter-underdrain-01	facility_underdrain	6 IN	38.00	0.53%	441.20->441.00	ASSUMED / GENERATED
storm-edge-outlet-01	private_storm_main	10 IN	12.21	0.82%	439.05->438.95	ASSUMED / GENERATED
storm-edge-public-connection-01	private_to_public_storm_connection	10 IN	19.12	1.05%	435.00->434.80	ASSUMED / GENERATED
roof-edge-leader-north	roof_leader	4 IN	7.00	1.43%	442.50->442.40	ASSUMED / GENERATED
roof-edge-leader-south	roof_leader	4 IN	57.00	1.05%	443.00->442.40	ASSUMED / GENERATED

LINED PLANTER TEST-BASIS DETAIL (NOT A CONSTRUCTION DETAIL)

1.25 FT DECLARED SURFACE STORAGE = 500 CF
FILTER MEDIA / DRAIN ROCK: FINAL SECTION UNKNOWN
LINER / 6 IN PERFORATED UNDERDRAIN: CONCEPTUAL

REQUIRED REVIEW / HOLD POINTS

1. Survey boundary, structures, rims/inverts, final grades, and benchmark.
2. Confirm architectural roof drains, overflow paths, and plumbing sizes.
3. Perform geotechnical infiltration/groundwater testing; no infiltration credit now.
4. Route hydrographs and size treatment/flow control/flood overflow.
5. Obtain City capacity, HGL, tie-in/outfall, material, and detail acceptance.
6. Pothole and survey sanitary/storm crossing; maintain accepted separation.
7. Inspect bedding, backfill, compaction, liner/media, testing, and as-builts.

KNOWN-ANSWER TEST CHECKS

Roof leaders connect through permanent terminal penetration-roof-drainage.

Every proposed storm segment falls downstream and meets its declared minimum slope/cover.

FC-1 internal drop = 1.95 FT; DMH-1 external drop = 3.95 FT.

Storm/sanitary modeled vertical clearance = 1.252694 FT (reviewed assumption).

PDF / GeoPackage / semantic package share stable IDs and geometry within 0.01 FT.

Public capacity, HGL, tie/outfall approval, survey, infiltration, and compliance remain UNKNOWN.

MODEL STUDIO - HILYARD DOMESTIC WATER / FIRE SERVICE PLAN

Separate fictional pressure networks from reference GIS context; no pressure, flow, capacity, hydrant-test, tie, meter, backflow, or Fire Marshal approval claim

REFERENCE / DESIGN BASIS

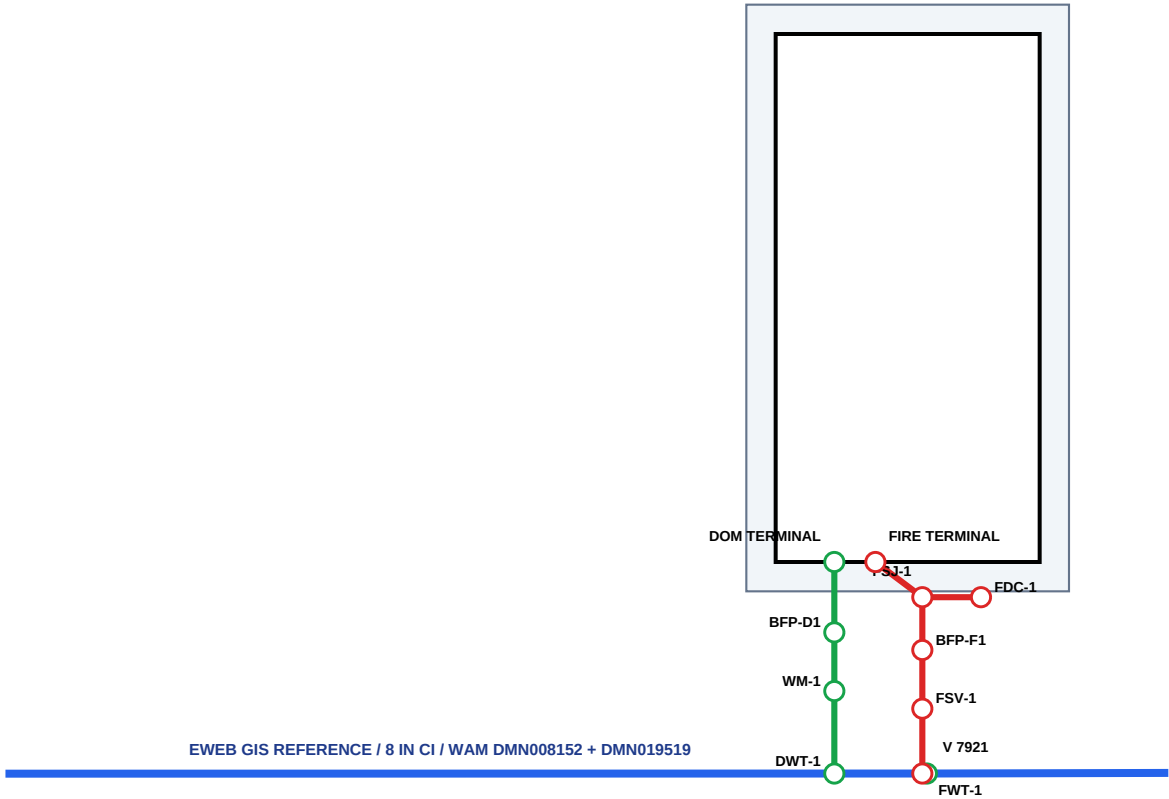
Fronting context: active 8 IN CI EWEB GIS main
GIS reference only - NOT SURVEY OR TIE AUTHORITY
Domestic: 2 IN HDPE DR11 / master meter / RPBA
Fire: 6 IN C900 DR18 / valve / detector double check
FDC branch: 4 IN assumption to FSJ-1
Modeled cover: 3.50 FT / declared minimum 3.00 FT
Pipe elevations and final grades: UNKNOWN
Pressure, residual pressure, capacity: UNKNOWN
Hydrant flow test and hydraulic model: UNKNOWN
Meter/backflow/tie/Fire Marshal approval: UNKNOWN

PLAN-SEPARATION SCREEN

Domestic to sanitary: 62.80 FT / 10.00 FT minimum
Domestic to storm: 39.36 FT / 2.00 FT minimum
Fire to sanitary: 64.29 FT / 10.00 FT minimum
Fire to storm: 33.29 FT / 2.00 FT minimum
All values are horizontal plan distance only
Vertical separation: UNKNOWN - pothole/survey required
No water line crosses a modeled storm facility

FIELD HOLD POINTS

- 1 Obtain EWEB pressure/capacity and approved tie basis
- 2 Calculate building demand and sprinkler/fire flow
- 3 Survey/locate/pothole utilities and final grades
- 4 Approve meter, backflow, FDC, valves, and restraint
- 5 Inspect tracer/bedding/restraint; pressure/disinfect/test
- 6 Capture certified tests and as-built coordinates/depths



MODEL STUDIO - HILYARD WATER / FIRE SEPARATION SCHEDULE / TEST DETAILS

Stable pressure-network IDs, horizontal coordination checks, restraint notes, field workflow, and explicit hydraulic/approval unknowns

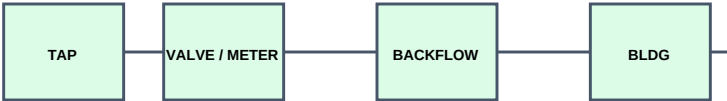
PRESSURE SERVICE EDGE SCHEDULE

EDGE ID	TYPE	DIA	LEN	COVER	MATERIAL	RESTRAINT	HYDRAULICS
dom-edge-tie-meter	private_domestic_service	2 IN	14.06	3.50 FT	HDPE DR11	manufacturer fused/re...	UNKNOWN
dom-edge-meter-backflow	private_domestic_service	2 IN	10.00	3.50 FT	HDPE DR11	fused service; device...	UNKNOWN
dom-edge-backflow-terminal	private_domestic_service	2 IN	12.00	3.50 FT	HDPE DR11	fused service with re...	UNKNOWN
fire-edge-tie-valve	private_fire_service	6 IN	11.06	3.50 FT	C900 PVC DR18	restrained joint/thru...	UNKNOWN
fire-edge-valve-backflow	private_fire_service	6 IN	10.00	3.50 FT	C900 PVC DR18	restrained joints at ...	UNKNOWN
fire-edge-backflow-junction	private_fire_service	6 IN	9.00	3.50 FT	C900 PVC DR18	restrained tee approa...	UNKNOWN
fire-edge-junction-terminal	private_fire_service	6 IN	10.00	3.50 FT	C900 PVC DR18	restrained tee and bu...	UNKNOWN
fire-edge-fdc-branch	fdc_branch	4 IN	10.00	3.50 FT	C900 PVC DR18	restrained FDC check-...	UNKNOWN

HORIZONTAL SEPARATION / VERTICAL-STATUS SCHEDULE

RELATIONSHIP ID	SYSTEMS	PLAN CLEAR	MINIMUM	VERTICAL	STATUS
domestic-sanitary-separation-01	domestic_water / sanitary	62.80 FT	10.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY
domestic-storm-separation-01	domestic_water / storm	39.36 FT	2.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY
fire-sanitary-separation-01	fire_water / sanitary	64.29 FT	10.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY
fire-storm-separation-01	fire_water / storm	33.29 FT	2.00 FT	UNKNOWN	PLAN PASS / FIELD VERIFY

CONCEPTUAL DEVICE / RESTRAINT DETAIL - NOT A CONSTRUCTION DETAIL



Continuous tracer/warning system; bedding/backfill/compaction per accepted details.

Restrain valves, tees, bends, device transitions, and building entry from approved thrust analysis.

Exact devices, vault/box drainage, freeze protection, and final grade/elevation remain UNKNOWN.

UNRESOLVED DESIGN / APPROVAL GATES

1. Building occupancy, fixture demand, irrigation/common-use demand
2. Required fire flow, sprinkler demand, hose allowance, duration
3. EWEB static/residual pressure, capacity, system impact, tie location
4. Hydrant flow test and water/fire hydraulic calculations
5. Meter and backflow hazard/device/placement approval
6. FDC/riser/access/signage and Fire Marshal approval
7. Pipe centerline elevations, final grades, crossings, thrust/restraint
8. Survey, permits, testing, disinfection, inspection, and acceptance